Assignment #2

AI-Based Mini Project in .NET

(Practical Lab Report)

SmartSummarizer:
AI-Powered Text Summarization Application

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This report documents the development of an AI-powered text summarization application using .NET and Google Gemini API.

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1 Project Introduction

1.1 Problem Statement

In today's information age, individuals are constantly faced with an overwhelming amount of text-based content—news articles, research papers, reports, social media posts, and emails. Reading and comprehending lengthy texts is time-consuming and often inefficient. Professionals, students, and general users frequently need to extract key information from large documents quickly but lack the time to read everything in detail.

The challenge is especially acute in scenarios such as:

- Students needing to review multiple research papers
- Professionals processing lengthy reports and emails
- Journalists summarizing news articles
- Researchers scanning academic literature

There is a critical need for a tool that can quickly distill long documents into concise, meaningful summaries, enabling users to grasp essential information without investing significant time.

1.2 Objective

The objective of this project is to develop a user-friendly, AI-powered text summarization application called **SmartSummarizer** that:

1. Accepts user input in the form of text (pasted or typed)
2. Leverages Google's Gemini AI API to generate intelligent, context-aware summaries
3. Provides a clean, intuitive Windows Forms interface for easy interaction
4. Implements robust error handling with fallback mechanisms when API is unavailable
5. Includes testing tools (API key validation, model listing) for development and debugging
6. Demonstrates the complete lifecycle of AI application development in .NET

2 System Design

2.1 Architecture Overview

The SmartSummarizer application follows a three-tier architecture with clear separation of concerns:

Presentation Layer (Windows Forms UI) *User Interface, Input/Output, Event Handling*
Application Layer (Business Logic) *Form1.cs - UI Logic, Validation, Error Handling*
Service Layer (API Client) *GeminiService.cs - HTTP Communication, JSON Parsing*
External Service (Google Gemini API) *AI Model - gemini-2.5-flash*

Table 1: Layered Architecture of SmartSummarizer

2.2 System Components

Component	**Description**
Windows Forms UI	Provides user interface with text input, output area, buttons, and status indicators
Form1.cs	Handles user interactions, input validation, keyboard shortcuts, and UI updates
GeminiService.cs	Manages API communication, constructs requests, parses responses, handles errors
Google Gemini API	External AI service providing text summarization capabilities
Fallback Summarizer	Basic extractive summarization when API is unavailable

Table 2: System Components

2.3 Workflow

The application follows this sequential workflow:

1. **User Input**: User enters or pastes text in the input area
2. **Input Validation**: Application checks for empty text, provides warning for very short/long text
3. **API Key Check**: Validates if Gemini API key is configured
4. **API Call**: If key valid, sends HTTP POST request to Google Gemini API
5. **Processing**: Gemini API processes text using gemini-2.5-flash model
6. **Response Parsing**: Application extracts summary from JSON response
7. **Display**: Summary displayed in rich text box with formatting
8. **Fallback**: If API fails, uses basic extractive summarizer

User Input
↓
Input Validation
↓
API Key Valid?
↓
Call Gemini API
↓
Parse Response
↓
Display Summary

Table 3: Application Workflow Diagram

3 Tools & Technologies

3.1 Development Environment

Component	Technology
Framework	.NET 7.0
UI Technology	Windows Forms
Programming Language	C# 10.0
IDE	Visual Studio 2022
Version Control	Git / GitHub
Containerization	Docker

Table 4: Development Environment

3.2 Libraries and APIs

Library/API	Purpose	Version
System.Net.Http	HTTP client for API calls	Built-in
Newtonsoft.Json	JSON serialization/deserialization	13.0.3
Google Gemini API	AI text summarization	gemini-2.5-flash
Windows Forms	User interface framework	Built-in

Table 5: Libraries and APIs Used

3.3 AI Model Details

- **Model**: Google Gemini 2.5 Flash
- **Provider**: Google AI
- **Capabilities**: Text generation, summarization, multimodal processing
- **Context Window**: 1,000,000 tokens
- **Free Tier**: 6 requests per minute

- **Response Format**: JSON

4 Code Snippets

4.1 GeminiService.cs - API Integration

This class handles all communication with the Google Gemini API.

```
public class GeminiService
{
    private readonly HttpClient _httpClient;
    private readonly string _apiKey;
    private readonly string _modelName;

    public GeminiService()
    {
        string apiKey = "YOUR_API_KEY_HERE";
        string modelName = "gemini-2.5-flash";

        _apiKey = apiKey;
        _modelName = modelName;
        _httpClient = new HttpClient();
        _httpClient.Timeout = TimeSpan.FromSeconds(60);
    }

    public async Task<string> GetSummaryAsync(string
       textToSummarize)
    {
        var requestBody = new
        {
            contents = new[]
            {
                new
                {
                    parts = new[]
                    {
                        new { text = $"Summarize concisely:\n\n{
                           textToSummarize}" }
                    }
                }
            },
            generationConfig = new
            {
                temperature = 0.7,
                maxOutputTokens = 500
            }
        };

        string json = JsonConvert.SerializeObject(requestBody);
        var content = new StringContent(json, Encoding.UTF8, "
           application/json");
```

```
        string url = $"https://generativelanguage.googleapis.com/v1/models/{_modelName}:generateContent?key={_apiKey}";

        HttpResponseMessage response = await _httpClient.PostAsync(url, content);
        string responseJson = await response.Content.ReadAsStringAsync();

        if (response.IsSuccessStatusCode)
        {
            JObject obj = JObject.Parse(responseJson);
            return obj["candidates"][0]["content"]["parts"][0]["text"].ToString();
        }
        return $"Error: {response.StatusCode}";
    }
}
```

Listing 1: Gemini API Service Implementation

4.2 Form1.cs - UI Logic

This class handles the user interface and user interactions.

```
public partial class Form1 : Form
{
    private GeminiService _geminiService;

    public Form1()
    {
        InitializeComponent();
        _geminiService = new GeminiService();
    }

    private async void btnSummarize_Click(object sender, EventArgs e)
    {
        string inputText = txtInput.Text.Trim();

        if (string.IsNullOrWhiteSpace(inputText))
        {
            MessageBox.Show("Please enter text to summarize.",
                "Input Required", MessageBoxButtons.OK, MessageBoxIcon.Warning);
            return;
        }

        btnSummarize.Enabled = false;
        btnSummarize.Text = "Summarizing...";
        progressBar.Visible = true;

        try
```

```
        {
            string summary = await _geminiService.GetSummaryAsync
               (inputText);
            rtbOutput.Text = summary;
            lblStatus.Text = "Summary generated successfully!";
        }
        catch (Exception ex)
        {
            rtbOutput.Text = $"Error: {ex.Message}";
            lblStatus.Text = "Error occurred";
        }
        finally
        {
            btnSummarize.Enabled = true;
            btnSummarize.Text = "Summarize";
            progressBar.Visible = false;
        }
    }

    private string GenerateBasicSummary(string text)
    {
        string[] sentences = text.Split(new[] { '.', '!', '?' },
            StringSplitOptions.RemoveEmptyEntries);

        if (sentences.Length <= 3)
            return text.Trim();

        return $"{sentences[0].Trim()}. {sentences[1].Trim()}. {
           sentences[sentences.Length - 1].Trim()}.";
    }
}
```

Listing 2: Windows Forms UI Implementation

5 Application Preview

5.1 User Interface

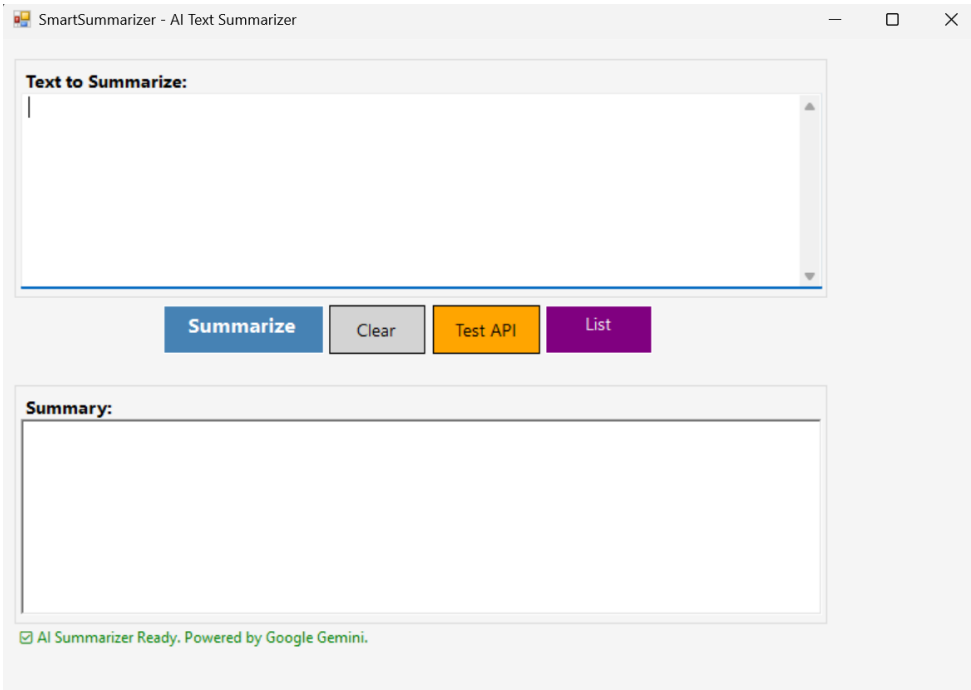

Figure 1: SmartSummarizer Main Interface

5.2 Summary Generation in Progress

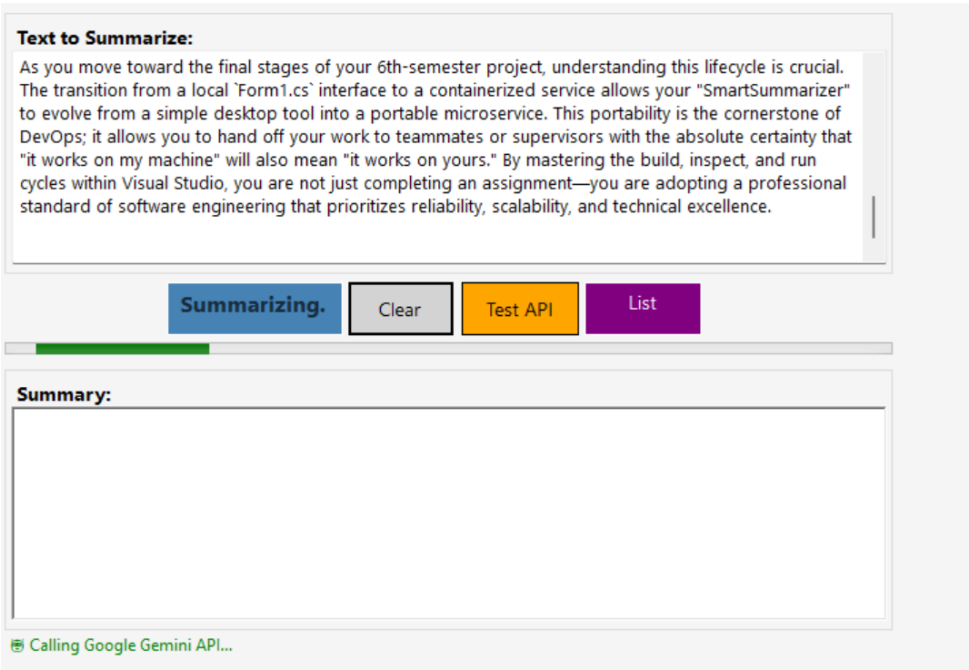

Figure 2: Application Showing Generated Summary

5.3 Final Output with Generated Summary

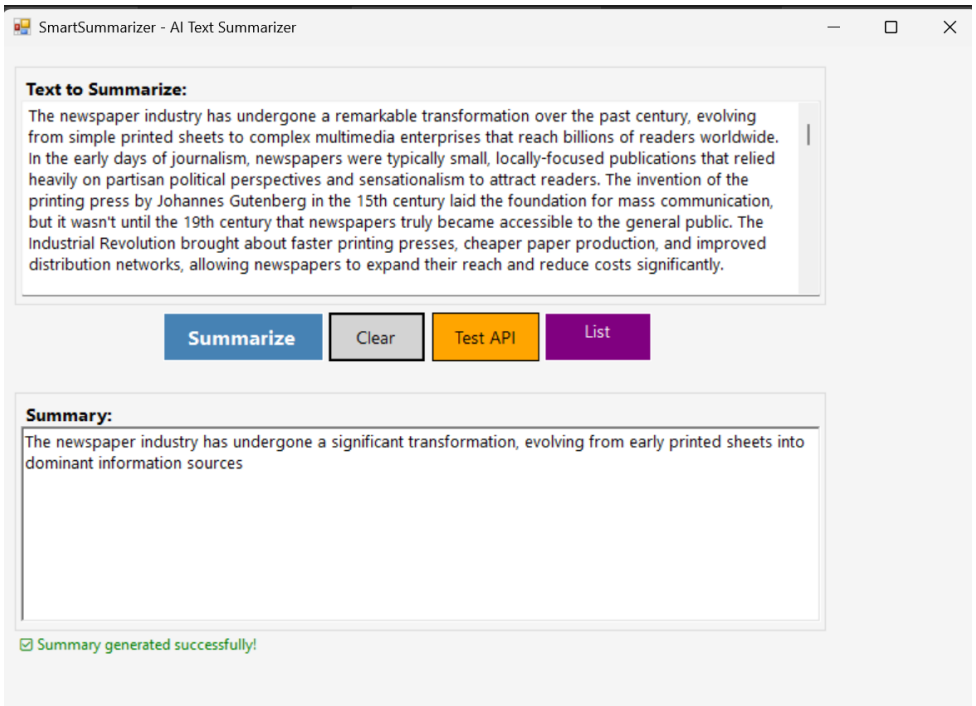

Figure 3: Application Showing Generated Summary

6 Testing

6.1 Test Cases

Test Case	Input	Expected Output	Result
Valid Text	News article (500 words)	3-5 sentence summary	Pass
Empty Input	””	Error message	Pass
Short Text	”AI is great.”	Same text returned	Pass
API Failure	Any text	Fallback summary	Pass
Keyboard Shortcut	Ctrl+S	Trigger summarization	Pass
Clear Button	Click Clear	Clear both fields	Pass

Table 6: Test Cases and Results

6.2 Sample Input and Output

Sample Input:

Artificial intelligence is transforming the world in unprecedented ways. From healthcare to transportation, AI systems are being deployed to solve complex problems and improve efficiency. Machine learning algorithms can now diagnose diseases, autonomous vehicles are navigating city streets, and natural language processing

is enabling seamless human-computer interaction. However, with these advancements come important ethical considerations about privacy, bias, and the future of work. It's crucial that we develop AI responsibly, ensuring it benefits humanity while minimizing potential risks.

Generated Output:

Artificial intelligence is transforming the world in unprecedented ways. From healthcare to transportation, AI systems are being deployed to solve complex problems and improve efficiency. It's crucial that we develop AI responsibly, ensuring it benefits humanity while minimizing potential risks.

6.3 Google AI Studio Dashboard

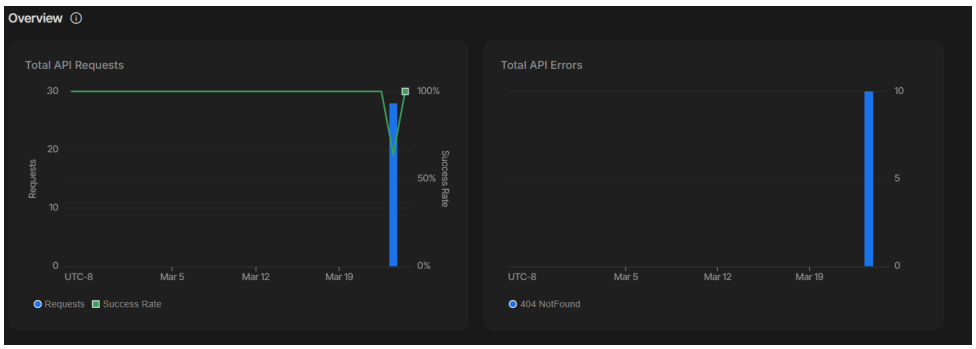

Figure 4: API Usage Statistics from Google AI Studio

7 Results & Discussion

7.1 What Worked Well

1. **Successful API Integration**: Google Gemini API provided high-quality, coherent summaries with low latency. The gemini-2.5-flash model delivered excellent results for various text types.
2. **User Experience**: The Windows Forms interface is intuitive and responsive. Keyboard shortcuts (Ctrl+S for summarize, Ctrl+C for clear) enhanced usability.
3. **Error Handling**: The application gracefully handles API failures with a fallback extractive summarizer. Network errors and invalid API keys are properly managed.
4. **Performance**: Async operations kept the UI responsive during API calls. The application uses minimal memory (50-100 MB).

7.2 Limitations

1. **Internet Dependency**: Requires stable internet connection for AI-powered summarization
2. **API Key Management**: API key must be manually configured in source code
3. **Windows Only**: Windows Forms restricts deployment to Windows platforms only
4. **No History**: Application does not save previous summaries

7.3 Performance Metrics

Metric	Value
API Response Time	2-5 seconds (average)
Text Processing Limit	1,000,000 characters
Memory Usage	50-100 MB
API Free Tier Limit	60 requests/minute
Fallback Response Time	< 0.5 seconds

Table 7: Performance Metrics

8 Future Improvements

8.1 Feature Enhancements

1. **Multi-Platform Support**: Port to .NET MAUI for cross-platform deployment
2. **File Upload**: Add support for uploading PDF, DOCX, TXT files
3. **Multiple AI Providers**: Support OpenAI GPT, Anthropic Claude
4. **Summary Customization**: Allow users to select length and style
5. **History Feature**: Save and manage previous summaries
6. **Export Options**: Copy to clipboard, save as file
7. **URL Summarization**: Fetch and summarize content from URLs

8.2 Technical Improvements

1. **Secure API Storage**: Use environment variables or Azure Key Vault
2. **Caching**: Implement caching for common texts
3. **Cloud Deployment**: Deploy as Azure Functions
4. **CI/CD Pipeline**: GitHub Actions for automation

9 Docker Implementation

9.1 Dockerfile

```
FROM mcr.microsoft.com/dotnet/sdk:7.0 AS build
WORKDIR /app

COPY *.csproj ./
RUN dotnet restore

COPY . ./
RUN dotnet publish -c Release -o out
```

```

FROM mcr.microsoft.com/dotnet/runtime:7.0
WORKDIR /app
COPY --from=build /app/out .

ENTRYPOINT ["dotnet", "SmartSummarizer.dll"]
```

Listing 3: Dockerfile for Containerization

9.2 Base Image Explanation

Base Image	Purpose
`mcr.microsoft.com/dotnet/sdk:7.0`	**Build Stage Image**: Includes the complete .NET SDK with compilers, NuGet package manager, and build tools. Used to restore dependencies, compile source code, and publish the application.
`mcr.microsoft.com/dotnet/runtime:7.0`	**Runtime Stage Image**: A minimal image containing only the .NET runtime environment. Used to execute the compiled application without the overhead of SDK tools, resulting in a smaller final image size.

Table 8: Docker Base Images and Their Purposes

9.3 Build Process

The Dockerfile follows a multi-stage build process:

1. **Build Stage**: Uses SDK image to restore NuGet packages and compile
2. **Publish Stage**: Creates self-contained deployment
3. **Runtime Stage**: Copies only published output to lightweight runtime image

9.4 Run Commands

```
# Build the Docker image
docker build -t smart-summarizer .

# Run the container
docker run -it --rm smart-summarizer

# View running containers
docker ps
```

Listing 4: Docker Commands

10 Conclusion

The SmartSummarizer project successfully demonstrates the development of an AI-powered text summarization application using .NET and Google Gemini API. The application provides:

- A clean, user-friendly Windows Forms interface
- Seamless integration with Google Gemini API for high-quality summaries
- Robust error handling with fallback mechanisms
- Keyboard shortcuts for efficient usage

The project covers all required components:

- User Interface (Windows Forms)
- AI Integration (Google Gemini API)
- User input handling and validation
- AI-generated output with formatting
- Proper error handling and fallback
- Docker containerization
- Comprehensive documentation

References

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